P-Set VI

POLI 784 - Spring 2026

2026-04-13

Complete all of the questions below together with your group members. Please submit both (1) a compiled PDF or HTML file with your responses and (2) the `R Markdown`, `Quarto`, etc. file containing both your responses and code as a group. You are encouraged to use `R Studio` projects and virtual environments to complete this p-set, although it is not required. Similarly, you are encouraged to use consistent style and linters when writing code.

Question 1

Consider the study in Sances (2016). Between 1980 and 2011, many New York towns changed the process of selecting real estate assessors from election to appointment. The study examines if appointed assessors ($D_i = 1$ – i.e., `appointed`) have stronger incentives to conduct reassessment (Y_i – i.e., `reassessed`). You can find the data needed to solve this exercise in `data1.csv`.

a. *(0.5 points)* Use `panelView()` to visualize treatment status for units in the dataset. Print the number of units (`swis_code`) and periods (`year`) in the sample to console.

b. *(0.5 points)* Create a new dataframe named `dat_DID` containing only units for which $cohort \in \{1, 14\}$, which gives you a dataset with the classic DID structure. Create indicators for the treatment group (i.e., `dat_DID$cohort == 14`) and post-treatment periods (i.e., `dat_DID$year >= 1999`), before applying the DID estimator to estimate the ATT and printing the result to console.

c. *(1 point)* Apply the two-way transformation to `dat_DID$reassessed` and `dat_DID$appointed` and fit an OLS regression using the transformed variables to estimate the ATT. How does the estimate differ from that in *1b* and should you worry about the issue of negative weights in this case? *(NOTE: For any variable Y_{it}, its two-way transformation equals $Y_{it} - \bar{Y}_{i.} - \bar{Y}_{.t} + \bar{Y}.$)*

d. *(0.5 points)* Use `feols()` to implement the within estimator for the data in `dat_DID` and ensure that the point estimate is the same as that in *1c*. Compare the OLS standard errors from this model to the clustered standard errors (clustered at the level of the unit) from a second model. What can you learn from the results?

e. *(0.5 points)* Plot the ATT estimate for each period with the leads-and-lags approach using the data in `dat_DID`. Do your results support the assumption of parallel trends?

f. *(0.5 point)* Apply `feols()` to the original dataset to obtain the TWFE estimate of the causal effect and print the point estimate to console. Should you worry about the issue of negative weighting in this case and why?

g. *(0.5 point)* Apply `feols()` to all untreated observations in the original dataset and use the resulting model to predict the counterfactual outcome for all treated observations. Estimate the ATT using the predicted counterfactual values and print the point estimate to console. *(NOTE: Your results will differ marginally from those in 1h due to the implementation of `feols()`, but you are applying the fixed effects counterfactual estimator manually.)*

h. *(1 point)* Use the fixed effects counterfactual estimator from `fect` to estimate the ATT in the original dataset. Print the ATT estimate to console and plot the ATT estimate for each period. Do your results support the assumptions underlying the estimator?

i. *(0.5 point)* Conduct a placebo test with three placebo periods and plot the results to validate the assumptions underlying the *FEct* estimator. Do your results support the assumptions?

Question 2

This exercise is based on Fouirnaies and Mutlu-Eren (2015), who investigated whether partisan alignment between local councils and the central government in England bring localities more grants. The outcome of interest is the logarithm of specific grants per capita allocated to a local council (i.e., `logSgwaPercap`). The treatment is a dummy variable indicating whether the incumbent party controls the majority of the local council (i.e., `treat`). You can find the data needed to solve this exercise in `data2.csv`.

a. *(0.5 points)* Use `panelView()` to visualize treatment status for the first 100 units in the dataset. Print the number of units (`councilnumber`) and periods (`year`) in the sample to console. Should we worry about the influence of carryover in this study?

b. *(1 point)* Use the fixed effects counterfactual estimator from `fect` to estimate the ATT.

i. Print the ATT point estimate to console and plot the ATT estimate for each period.
ii. Conduct a placebo test with three placebo periods and plot the results.
iii. Conduct a carryover test with three post-treatment periods and plot the results.
iv. Do your results support the assumptions underlying the estimator?

c. *(1 point)* Use the **interactive** fixed effects counterfactual estimator from `fect` to estimate the ATT. *(NOTE: Use the default number of factors.)*

i. Plot the ATT estimate for each period.
ii. Conduct a placebo test with three placebo periods and plot the results.
iii. Conduct a carryover test with three post-treatment periods and plot the results.
iv. Do your results support the assumptions underlying the estimator?

Question 3

(2 points) Use Cinelli and Hazlett's (2020) sensitivity analysis to test the robustness of Blattman and Annan's (2010) findings. You can find the data needed to solve this exercise in `data3.csv`.

i. Regress `educ` on all other variables in the dataset using `lm()` to estimate the ATE of `abd`. Use `age` as the benchmark covariate for the sensitivity analysis.
ii. Plot both the bias contour of the point estimate and the extreme scenario.
iii. Print the summary of the sensitivity analysis to console and interpret your findings.